

Received May 9, 2020, accepted June 2, 2020, date of publication June 22, 2020, date of current version July 20, 2020.

Digital Object Identifier 10.1109/ACCESS.2020.3002825

Robust Autonomous Intersection Control Approach for Connected Autonomous Vehicles

YUHENG ZHANG^{1,2}, LUNING LIU^{1,2}, ZHAOMING LU^{1,2}, (Member, IEEE),
LUHAN WANG^{1,2,3}, (Member, IEEE), AND XIANGMING WEN^{1,2}, (Senior Member, IEEE)

¹Beijing Key Laboratory of Network System Architecture and Convergence, Beijing 100088, China

²School of Information and Communication Engineering, Beijing University of Posts and Telecommunications, Beijing 100876, China

³Witcomm Open Source Communication Research Institute, Beijing 100876, China

Corresponding author: Zhaoming Lu (lzy0372@bupt.edu.cn)

This work was supported in part by the National Key Research and Development Program of China under Grant 2019YFB1803301, in part by NSFC Project under Grant 61801047, and in part by the Beijing Nova Program of Science and Technology under Grant Z191100001119028.

ABSTRACT Traffic light-free intersection control is envisioned to alleviate congestion and manage vehicles intelligently. With the help of vehicle-to-infrastructure (V2I) communication and edge computing (EC), vehicles are instructed to cross the intersection with high vehicle safety and traffic efficiency without traffic lights. However, unstable channel conditions can lead to the reduction of traveling safety. In this paper, we propose a robust autonomous intersection control (AIC) approach with global optimization scheduling, which protects connected autonomous vehicles from collision under any channel conditions while achieving decent traffic efficiency. In particular, we propose an AIC model that gives vehicles certain autonomy under centralized control to ensure the traveling safety in case of some emergencies. By conducting an interference graph, we simplify the AIC problem as a weighted maximal clique problem with restriction. To improve the fairness and efficiency in terms of vehicle passage, multiple factors such as travel delay, traffic of the current lane and passengers' desired speed are considered. Furthermore, we propose a heuristic algorithm to search the solution space. For further optimization, a particle swarm optimization algorithm is proposed, achieving a near-optimal result with adjustable overhead. Finally, we build the simulation model and conduct a comparative performance evaluation. Simulation results demonstrate the superiority of our proposed scheme.

INDEX TERMS Connected autonomous vehicles, autonomous intersection control, edge computing, robustness, vehicular network, optimization.

I. INTRODUCTION

Known as a bottleneck of traffic, intersections are the area that causes many accidents and traffic congestion [1]. Traffic statistics suggest that vehicle collisions at intersections accounted for between 25% and 45% of all collisions, most of which are caused by human errors [2]–[4]. And congestion caused people in urban areas to spend 6.9 billion hours more on the road [5]. Autonomous vehicles have always attracted researchers as it is widely recognized that they can not only improve road safety and relieve congestion but have the potentials to enhance transportation intelligence and provide in-vehicle entertainment experience. With the development in computer science, communication technology and sensing,

vehicles with varying levels of automation are already on the road, these technological advancements make autonomous intersection control (AIC) a feasible option to realize “Intelligent Transportation”.

In the conceived AIC system, autonomous vehicles connect with an agent deployed at the intersection by leveraging vehicle-to-everything (V2X) communication. The agent instructs vehicles to cross the intersection cooperatively according to their real-time kinematic states [6]. Different from traditional ad hoc networks, most nodes in V2X networks are vehicles with high mobility and different trajectories, which poses challenges to V2X communication. Efforts in various aspects have made to facilitate stability improvement and latency reduction. In terms of communication standards, C-V2X (Cellular Vehicle-to-Everything), which has 5G new radio features and predictable performance, has

The associate editor coordinating the review of this manuscript and approving it for publication was Celimuge Wu¹.

developed rapidly. C-V2X enables vehicles to collect information from various sources simultaneously and create a real-time map of the surroundings, ensuring AIC has enough information to make smarter and safer scheduling. In the research area, extensive studies have been done to improve link reliability and quality of service. In [7] and [8], V2X communication protocols with edge intelligence are proposed, which achieves high throughput and low latency and in highly dynamic V2X networks. With the current advancement of V2X, primary AIC, namely, vehicle speed guidance, has been tested in many places. The intersection informs the vehicle of the traffic light phase via V2X, and then the vehicle adopts a speed change strategy so that it can reach the stop line and pass through during the green light time to avoid a stop.

Considered as a more suitable solution for enhancing computational capabilities in vehicular environments [9], edge computing (EC) supports computing and decision services for AIC in the future intelligent transportation system (ITS). To meet the requirements of future Cooperative, Connected and Automated Mobility (CCAM), a part of traffic and computation needs to be pushed to the network edges [10]. Edge computing servers provide information service environment and cloud computing capability at the edge of the network, which supports real-time regional centralized services with low end-to-end latency [11]. In the future, other centralized computation/management services produced by various automated driving applications such as data aggregation, crowdsensing participant recruitment, and route guidance will be integrated to the edge for network simplicity, which will pose the challenge to the computing capacity of edge computing. Although many enhancements like offloading, caching mechanism [12], and service placement mechanism [13] have proposed to promote the adaptation of EC to V2X applications, a lightweight AIC scheme is preferable to make decisions in a timely manner.

Nowadays, most intersections are controlled with traffic lights or stop signs for collision avoidance, and the former is usually reserved for larger, busier intersections [14]. However, under the control of traffic lights, the traffic flow is cut simply and roughly. Furthermore, traffic lights have few signal light phases and cannot adjust the phase in real time according to the current situation. Although researchers have proposed many ways to improve the performance of traffic lights, it still has its limitations [15]–[18]. Nevertheless, the removal of traffic lights raises several challenges. Traffic lights are an effective intersection control mechanism that has been in use for more than 150 years. Their practicality have been widely recognized. As mentioned in [19], to build a new AIC system, several properties must be well considered.

In any AIC system, safety must be put in the first place. AIC mechanisms must require a reliable collision avoidance method. AIC system sends out instructions and exchanges vehicular kinematics status and road condition information through the network. But travelling safety should be guaranteed even when some vehicles are disconnected, which will

lead to information inconsistency between the vehicle and EC server. The solution proposed in this paper maintains the travelling safety under different channel conditions, which is seldom concerned in the previous research. In addition, the following properties are supposed to be satisfied:

- Efficiency. All vehicles should pass the intersection as soon as possible, which means that short waiting time and queue length of vehicles at the intersection are preferred.
- Autonomy for vehicles. Vehicles should maintain certain autonomous control capabilities, so that they can respond to deadly hazards even if they are not in accordance with instructions directed by EC server.

Considering the above properties, in this paper, we propose an robust reservation-based AIC scheme for autonomous vehicles. First, we analyze the AIC problem, and formulate it as a objective function in a moderate granularity. Then, we propose a heuristic method to solve this NP-hard problem. It is worth noting that, only connected autonomous vehicles (CAVs) are considered in this paper. The main contributions of this paper are summarized as follows:

- 1) We propose an edge-assisted traffic control architecture for CAVs. We divide intersection into multiple collision areas, and define the trajectories that vehicles in different motion (turning left, turning right or going straight) pass through for model simplification. Vehicles drive under the guidance of EC server. With our proposed scheme, robustness is guaranteed under all channel conditions.
- 2) We formulate the traffic control model as a weighted maximal clique problem. All the possible collisions are predicted, and then are abstracted with a weighted interference graph. Besides, multiple characteristics such as travel delay, current lane traffic and passengers' desired speed are taken into account for system fairness and traffic efficiency consideration.
- 3) We propose a stack-based tabu search algorithm to search the solution space, according to the weighted interference graph. To improve the efficiency of resource allocation, a particle swarm optimization (PSO) algorithm is implemented for further performance optimization. With the proposed algorithms, the collision can be avoided no matter which vehicles are disconnected at any time.
- 4) We evaluate the performance of the scheme. Compared with other management algorithms and traffic control rules, the superiority of our algorithms in transportation efficiency and robustness is demonstrated.

The rest of the paper is organized as follows. Section II reviews the related work. Section III presents the system model and formulates the traffic control model. Section IV provides AIC solutions by using heuristic algorithms. Section V performs the simulation results. Section VI concludes this paper and discusses future research directions.

II. RELATED WORK

Over the last decade, there is a rich literature dedicated to promoting traffic efficiency of intersection on the basis of traffic signal control. Most of traffic signal control systems present adaptive mechanisms using various techniques such as particle swarm optimization [15], reinforcement learning [16], [17] and evolutionary algorithms [18]. These algorithms have achieved good performance in the traffic light intersection with variable traffic arrival rate. However, if the traffic arrival rate is fixed, or the intersection is busy, their optimizations do not have much effect.

Traffic signal control methods optimized the duration and combinations of traffic phases depending on the real traffic conditions. However, they are unsuitable for autonomous vehicles owing to the overly conservative pattern of fixed phrases. Thus, researchers have proposed a number of solutions for non-connected autonomous vehicles. For example, the method proposed in Ref. [20] ensures the safety of autonomous vehicles and safeguards overly conservative by combining motion prediction, threat assessment, and decision-making methods. With the introduction of vehicular communication networks, information collected by vehicles and infrastructure are capable of being exchanged through the network [21].

At the same time, researchers have conducted a lot of research on the vehicular Internet of Things. It is particularly challenging to achieve a high throughput and low latency with limited wireless resources in V2X communication and achieve a high processing speed and low latency in edge computing. To enhance V2X communications, Ref. [7], [8] provide “vehicle as an edge” solutions. With the intelligence with fuzzy logic and reinforcement learning, vehicles are classified as hierarchical nodes. In order to guarantee the sufficient computational capacity, an edge computing node can offload its computing tasks generated by the in-vehicle IoT application, researchers are trying to explore all usable computing devices. All vehicles, RSUs and the cloud can be developed as potential data caching nodes and computing nodes according to Ref. [9], [11]–[13], and the task volume that a node is assigned is reasonably allocated according to its computational capacity, computing speed and transmission delay, etc. The development of V2X communication and edge computing provides a good support for connected autonomous driving, and it encourages the emergence of researches on non-signalized autonomous intersection control (AIC).

Generally, AIC has two different approaches. One approach is to assign each vehicle’s velocity through the whole travel, called velocity assignment approach. Li and Wang [22] early proposed a cooperative driving scheme at intersections without traffic lights. To represent the collision-free movements of vehicles at the intersection, a concept of safety driving patterns was introduced. The solution space of all movements was described by a spanning tree and then trajectory planning algorithms were used to determine the driving plans. To reduce the computational

complexity, Ref. [23], [24] adopted greedy first-in-first-out rule to find the solution. In [23], Chouhan *et al.* designed fixed trajectories for vehicles, and planning collision avoidance only in the all possible collision points, which reduces computational effort a lot. In [24], He *et al.* proposed a novel approach to erase on-road lane change by allowing the vehicle to turn onto any downstream lane at the intersection. Then in [25], [26], quality-of-experience (QoE) was taken into consideration. In [25], Krajewski *et al.* formulated the traffic as a graph, and found a solution keeping the balance of passengers’ driving comfort of global optimization. Later in [26], Dai *et al.* proposed a QoE-oriented approach using convex optimization. However, as driving comfort is optimized, the instruction of the coordinator of velocity and acceleration at each moment should be perfectly executed by the vehicle, which brings potential safety hazards. In [27], Lamouik *et al.* proposed a deep reinforcement learning-based AIC system with the ability to adapt to any intersection architecture quickly. The deep Q-network used vehicle information to output vehicle actions (keep speed, accelerate, or decelerate) to adjust the speed of vehicles in real-time. But as proofed in [19], the velocity assignment intersection control is not robust against model mismatches and external disturbances.

In another way, researchers proposed reservation-based AIC. In [14], distributed control is applied in the system where the intersection is divided into space-time resource blocks. Vehicles at the intersection should apply for these space-time resources. In order to avoid the collision, they adopted the first-in-first-service strategy. The simulation shew the superiority of this mechanism in traffic efficiency compared with traffic lights and stop signs. Later, they proposed a multi-intersection management approach in [28]. In [29], an approach combining with velocity assignment and reservation management was proposed. The area of intersection was discretized to nodes, and dynamic programming was implemented for the node reservation of vehicles. However, it cannot be robust against mismatches and external disturbances owing to its velocity assignment mechanism. Shi *et al.* [6] proposed a fully distributed algorithm employing 5G-V2X technology for industrial vehicles. Vehicles continue to collaborate with each other to set a timetable and then pass through with minimal delay. In addition to ensuring collision-free between vehicles, pedestrian safety was also taken into consideration. Sayin *et al.* [30] proposed an information-driven AIC to enable the agent to consider driver-exclusive information reported by drivers to enhance the quality of transportation. To avoid the misreporting of the driver-exclusive information, a payment-based incentive and compatible mechanism which guarantees truthful utility and maximizes the social welfare was designed. Choi *et al.* [31] designed a robust AIC approach which allows the vehicles to travel in a speed range to make the model applicable to the real-world system. However, its longer reservation time led to lower utilization. As an extension from that work, Choi *et al.* [32] proposed a scheme wherein collision can still be avoided when the vehicle fails to fol-

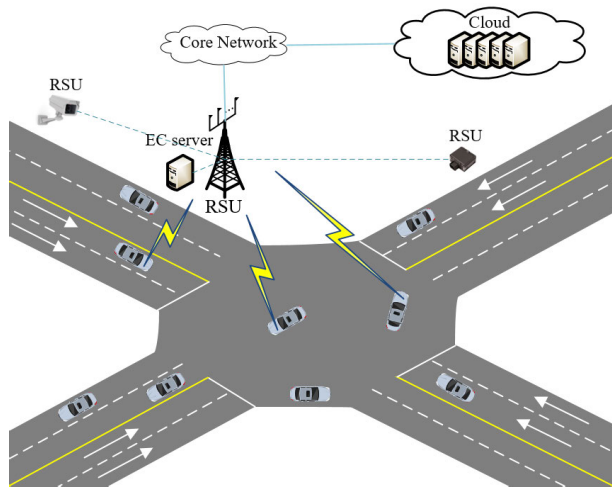


FIGURE 1. The architecture of the AIC system.

low the required speed range by instructing all vehicles to stop safely. Sahba *et al.* [33] addressed the AIC problem in the form of a network utility maximization problem to minimize the computational delay of algorithms. Miculescu and Karaman [34] introduced a polling mechanism to limit maximum expected delay, and consequently guaranteed the performance. Through the analysis of the queuing system, the stability of the algorithm was proved.

As far as we know, the only algorithm considering the influence of channel condition in the AIC system is proposed in Ref. [25]. However, when the channel condition is unstable, its inconsistency of information between EC server and vehicles can result in collisions that the EC server cannot predict.

Overall, due to the lack of flexibility, the performance of traffic signal control is limited. On the other hand, previous work on AIC have weakness in coping with unstable channel conditions and external disturbances. Our AIC solution considers the influence of unstable channel conditions and external disturbances to ensure its robustness while presenting near-optimal performance in terms of traffic efficiency. Furthermore, multiple factors such as queuing length, waiting time and desired speed of individual vehicles are considered for system fairness. Last but not least, our proposed solution enables adjustable computational overhead for EC servers with different computational capacity.

III. SYSTEM MODEL

A. SYSTEM OVERVIEW

In this paper, we adopt an edge cloud-enabled framework, which is described in [35]. Fig. 1 depicts the network topology in a four-way intersection. Roadside cameras and roadside radars are connected to the network to acquire more sensing data. Together with the base station, they are called the roadside units (RSUs). EC servers are deployed near the base station. By implementing EC servers, the computation in vehicles can be offloaded to the edge of networks.

When vehicles enter the management region (the region where AIC needs to pay attention to the status of vehicles for decision-making), they provide the following basic information to the EC server:

- Vehicle ID;
- Maximum acceleration and deceleration a_{max} and a_{min} ;
- Vehicle length L ;
- Current lane ID;
- action at intersection (including turning left, turning right or going straight);
- The desired speed v^d .

Here we assume that all vehicles have a desired speed v^d , and all vehicles want to be able to drive at their desired speed. v^d is determined based on the road speed limit, traffic volume, and passenger behavior preferences and can be changed through long-term driving data experience or different driving modes that the passenger previously sets up (e.g. comfort or speed priority). Such preferences can help passengers get a better travel experience.

Then, vehicles and RSUs observe perceptual information which including kinematic information (e.g., vehicle positions, speed, acceleration/deceleration, etc.) and pavement information (e.g., pedestrians, animals or obstacles, etc.), and report them to the EC server periodically. Among them, the real-time kinematic information of vehicles (including current position, speed and acceleration/deceleration) is required for the EC server for decision-making. To keep the timeliness and reliability of information, The link between the server and the vehicle does not adopt a vehicle as a relay, i.e., only vehicle-to-infrastructure (V2I) communication was used for information transfer.

Based on the information collected, the EC server is in charge of computing the collection of vehicles allowed at intersection. Then, the EC server distributes instruction messages to permit vehicles through V2I. The instruction message including the following information:

- Passing permission;
- Adjusted desired speed v^d .

Fig. 2 depicts the information exchange process after the vehicle enters the area. The EC server periodically computes the solution with a fixed schedule interval and then sends an instruction message to each permitted vehicle according to the calculation result. If the vehicle does not have permission to pass, it will not receive any messages. The vehicle keeps listening to messages from the EC server and it must stop before the stop line until it receives the instruction message. Once the vehicle gets the permission, its motion must follow the predefined rules to make sure that its behavior is predictable; otherwise, the permission will be withdrawn.

To ensure a collision-free travel of vehicles in the intersection, we consider a reservation-based AIC framework. The intersection is divided into collision blocks and some blocks will be occupied when the vehicle is driving in the intersection. A resource occupation table that records the dwelling interval of vehicles in CBs is maintained in the EC

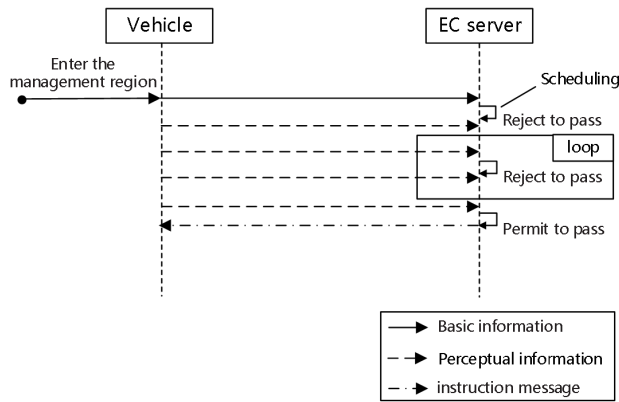


FIGURE 2. A diagram of a vehicle uploading information and EC issuing the instruction.

server. In this table, the occupancy of each CB is stored in a red-black tree in order of arrival time to quickly add and find elements [31]. Fig. 3 shows the workflow of the EC server calculating process. The procedures of the EC server can be divided into three parts. First, the EC server updates traffic information in the whole jurisdiction. Second, the EC server provides the solution for vehicles. Finally, the EC server sends the instruction to permitted vehicles. Vehicles has its autonomy to deal with unpredictable emergencies (such as the vehicle's disconnection and the appearance of unconnected obstacle). When the movement of the vehicle is contrary to the instructions of the EC server, which means an unpredicted emergency occurred, the EC server needs to re-compute the solution. As for dealing with disconnection, when a vehicle is disconnected unexpectedly, then a) if there is a permission for it, then it moves as the EC server instructs; b) if the vehicle cannot follow the instruction of the EC server or there is not a permission for it, then it stop before the stop line. In this way, safety is maintained even in case of emergencies. When the vehicle passes the intersection and reaches the departure region (the region where vehicles return to normal driving), the AIC system stops monitoring the vehicle, and the desired speed of the vehicle returns to the value before the approaching the intersection.

To focus on the design of traffic control system, in this paper, we make the following assumptions to simplify the traffic environment without losing the integral factors of the problem (i.e., efficiency, robustness and autonomy):

- 1) All the vehicles are CAVs. Vehicles are connected to the network. They are able to receive instructions from EC server and drive follow the instruction. If a vehicle is disconnected, it act according to the last instruction it receives.
- 2) Vehicles turning at the intersection follow a fixed trajectory. When passing the intersection, vehicles is able to follow the specified trajectory that EC server instructs.

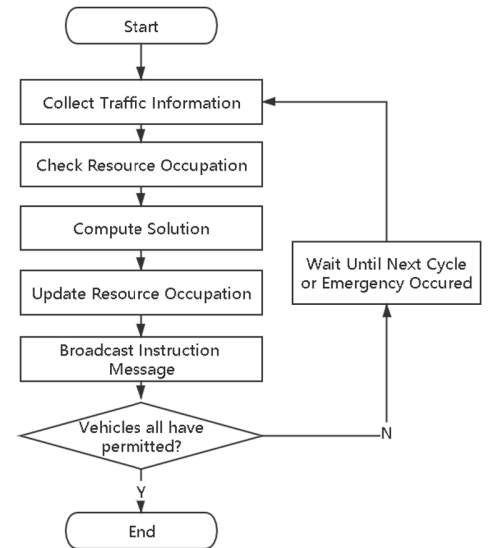


FIGURE 3. Work-flow of the EC server.

- 3) The transmission time for vehicular information to the EC server is in milliseconds [19]. Hence, we neglect the change of vehicular movement in the process of information transmission.
- 4) Pedestrians are not considered. The reserved time for pedestrians can be adjusted according to the real situation, but it is not considered in this paper.

Table 1 lists notations given to different parameters in this work.

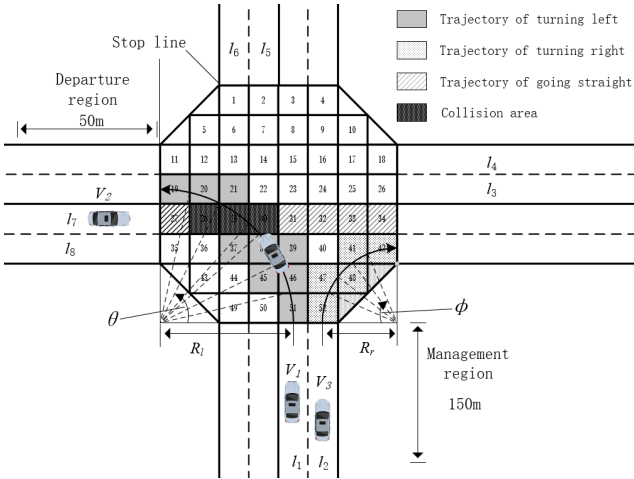
B. OBJECTIVE

AIC can be regarded as a resource allocation problem, in which the time and space of the intersection is the resource to be requested by the vehicles. The task is to reserve the intersection resources for vehicles, so that no more than one vehicle occupies the same space at the same time. As proved in [36], scheduling vehicles crossing an intersection is an NP-Hard problem. At the same time, AIC is time-critical. This contradiction motivates us to simplify the model and present a heuristic approach.

To simplify the model without loss of analytical precision, we divide the intersection into collision blocks (CBs) whose side width is the width of a lane (52 CBs for an ordinary four-way intersection). An example of an ordinary four-way intersection is shown in Fig. 4. Each lane is denoted with an ID from l_1 - l_8 , in which vehicles are allowed to turn left and go straight in the inner lane, whereas turn right and go straight in the outer lane. While entering the intersection, the vehicle follows a fixed trajectory starting from the centerline of the lane, and the trajectory overlaps with some CBs. We assume that the left-turning vehicle travels along a quarter circle with a radius of 4.5 times the width of the lane, which is denoted as R_l , and the right-turning vehicle travels along a quarter of a circle with a radius of 2.5 times the width of the lane, which is denoted as R_r .

TABLE 1. Summary of Notations.

Notation	Description
cb_i	The i th collision block
W	The width of a lane
l_k	The k th lane at intersection
T_i	The set of collision blocks which the trajectory of V_i overlays
$T_{i,n}$	The n th elements in T_i
$I_{i,k}$	The intersection of T_i and T_j
L_i	The length of vehicle V_i
$t_{i,k}^e$	The time when V_i entering the collision block cb_k
$t_{i,k}^l$	The time when V_i leaving the collision block cb_k
t_0	The current time
d_i	The travel distance of V_i
a_i^n	The normal acceleration and deceleration set for V_i
a_i	The current acceleration of V_i
v_i	The current speed of V_i
X_i	The current position vector of V_i
sa_1	The minimum safe distance for V_i to not slow down
sa_2	The minimum safe distance for V_i to not emergency brake
v_i^d	The desired speed of V_i
s_i^l	The distance from the current position of V_i to the stop line
$s_{i,n}^l$	The distance from the stop line to the entry $T_{i,n}$
$\mu_i(t)$	The vertex weights of V_i in the interference graph

**FIGURE 4.** Simplified system model with CBs and some parameters.

We mark the CBs as $cb_1 - cb_{52}$. Let V_i represents vehicle i . Denote T_i as the set of collision blocks which the trajectory of V_i overlays, and $I_{i,j}$ as collision area of T_i and T_j . As V_1 and V_2 are respectively in l_1 and l_7 , $T_1 = \{cb_{51}, cb_{46}, cb_{45}, cb_{39}, cb_{38}, cb_{37}, cb_{30}, cb_{29}, cb_{28}, cb_{21}, cb_{20}, cb_{19}\}$ and $I_{1,2} = T_1 \cap T_2 = \{cb_{28}, cb_{29}, cb_{30}\}$. The value of the elements in the set T_i is arranged according to CBs' order in which the vehicle V_i passes through. For instance, in set T_i , $cb_{39} > cb_{38}$. We label the CBs in the set according to the order of passing through CBs by assigning the elements $T_{i,1}, T_{i,2}, \dots, T_{i,\|T_i\|}$ in the set T_i in declining order. For the convenience of explanation and without loss of generality, our subsequent analysis is carried out under the intersection depicted as Fig. 4.

The areas leaving the intersection for all directions are the departure region, and the areas approaching the intersection for all directions are the management region, which is 150 m according to Ref. [23]. For vehicles in the management region, the EC server arranges three-dimensional (two for space and one for time) resource blocks for vehicle

trajectories achieving optimal resource utilization, which is similar to the three-dimensional Knapsack problem. Assume that vehicle V_i enters cb_k at time $t_{i,k}^e$ and leave at time $t_{i,k}^l$, denote the current time as t_0 and the travel distance of V_i as d_i . To make vehicles pass the intersection as soon as possible, we maximize the average travel speed of all vehicles. Considering $t_{i,k}^e$ and $t_{i,k}^l$ as optimization variables, the optimization problem can be given by:

$$\max \frac{1}{N} \sum_{i=1}^N \left[d_i / \left(\sum_{k=1}^M \alpha_{i,k} (t_{i,k}^l - \beta_{i,k} t_{i,k}^e) - t_0 \right) \right], \quad (1)$$

where N, M respectively represents current scheduling vehicle number of vehicles and total number of CBs, and $\alpha_{i,k}$ and $\beta_{i,k}$ is defined as:

$$\alpha_{i,k} = \begin{cases} 1, & cb_k \in T_i \\ 0, & cb_k \notin T_i, \end{cases} \quad (2)$$

$$\beta_{i,k} = \begin{cases} 1, & cb_k = T_{i,1} \\ 0, & cb_k \neq T_{i,1}. \end{cases} \quad (3)$$

Constraints of the AIC problem are discussed below. Denote a_i as the acceleration of V_i , and a_{imin} and a_{imax} respectively represent the maximum deceleration and acceleration of V_i . The following constraints should be satisfied:

$$a_{imin} \leq a_i(t) \leq a_{imax}, \quad (4)$$

$$0 \leq v_i(t) \leq v_{max}, \quad (5)$$

where

$$\int_t^{t_0} a_i(t) dt = v_i(t). \quad (6)$$

To make sure that the two vehicles do not collide, it has to be satisfied that for any cb_k , there is a buffer interval between the dwelling interval for all the vehicles:

$$t_{i,k}^e - t_{j,k}^l > 0 \quad \vee \quad t_{j,k}^e - t_{i,k}^l > 0, \quad \forall i \neq j, i, j \text{ not in the same lane}, \quad (7)$$

$$t_{j,k}^e - t_{i,k}^l > 0, \quad \forall i < j, i, j \text{ in the same lane}, \quad (8)$$

Constraints of $t_{i,k}^e$ and $t_{j,k}^l$ are given according to the kinematics formula. Use L_i to represent the length of V_i , and c_i to represent the curve length of V_i 's trajectory that passes through cb_k . Denote s_i^l and $s_{i,n}^l$ as the distance from the current position of V_i to the stop line and the distance from the stop line to the entry of $T_{i,n}$. we can obtain that:

$$\int_{t_0}^{t_{i,n}^e} v_i(t) dt = s_i^l + s_{i,n}^l, \quad (9)$$

$$\int_{t_{i,m}^e}^{t_{i,k}^l} v_i(t) dt = L_i. \quad (10)$$

In constraint (10), cb_m represents the next CB that vehicle V_i passes after passing cb_k , that is:

$$m = \arg \max_m cb_m, \quad cb_m, cb_k \in T_i \wedge cb_m < cb_k. \quad (11)$$

TABLE 2. Correspondence between the incoming $T_{i,n}$ and steering angle.

$T_{i,n}$	θ	value	$T_{i,n}$	ϕ	value
$T_{i,1}$	θ_0	0.0°	$T_{i,1}$	ϕ_0	0.0°
$T_{i,2}$	θ_1	12.9°	$T_{i,2}$	ϕ_1	23.1°
$T_{i,3}$	θ_2	12.9°	$T_{i,3}$	ϕ_2	37.5°
$T_{i,4}$	θ_3	26.5°	$T_{i,4}$	ϕ_3	53.0°
$T_{i,5}$	θ_4	26.5°	$T_{i,5}$	ϕ_4	66.9°
$T_{i,6}$	θ_5	41.7°	$T_{i,6}$	ϕ_5	90.0°
$T_{i,7}$	θ_6	41.7°			
$T_{i,8}$	θ_7	48.2°			
$T_{i,9}$	θ_8	63.4°			
$T_{i,10}$	θ_9	63.4°			
$T_{i,11}$	θ_{10}	63.4°			
$T_{i,12}$	θ_{11}	76.9°			
$T_{i,13}$	θ_{12}	90.0°			

where $\theta_0 - \theta_7$ is defined for left turning vehicles, whereas $\phi_0 - \phi_5$ is defined for right turning vehicles.

Particularly, if cb_k is already the minimum in T_i , $m = 0$. At this time, $t_{i,m}^1$ indicates the time when the vehicle V_i is ready to exit the intersection. The way to calculate $s_{i,n}^1$ in Constraint (10) is discussed below. For straight-through vehicles, $s_{i,n}^1$ can be obtain that:

$$s_{i,n}^1 = (n - 1)W. \quad (12)$$

For turning vehicles, the arc length of turning trajectory needs to be considered to ensure the accuracy of the prediction. As shown in Fig. 4, the radius of the left and right turning trajectories are known, and it is necessary to obtain the turning angle into each $T_{i,n}$. In particular, in the trajectory of a left turn, some different $T_{i,n}$ may share the same turning angle for ease of computation without losing too much precision. Table 2 lists the different left and right turn trajectories in the case of Fig. 4.

Therefore, the complete expression of $s_{i,n}^1$ is written as follows:

$$s_{i,n}^1 = \begin{cases} (n - 1)W, & \text{straight,} \\ R_l \theta_{n-1}, & \text{turning left,} \\ R_r \phi_{n-1}, & \text{turning right.} \end{cases} \quad (13)$$

IV. METHODOLOGY

As discussed in [19], centralized control schemes are weak to resist external disturbances. Therefore, we adopt a semi-centralized scheme in which vehicles have certain autonomy for action, and the EC server has the authority to manage passing permissions and to change the desired speed of vehicles. Noticing that constraints (6), (9), (10) are variable-limit integral formulas that are difficult to compute, it motivates us to adopt a heuristic method to obtain an optimized solution within a controlled calculation time. In this section, a PSO-based semi-centralized solution is proposed. The proposed solution enables that the EC server changes the desired speed and indirectly adjusts the speed of vehicles. This section is organized as follows. In subsection A and subsection B, collision prediction and collision avoidance mechanism of the proposed scheme is introduced. In subsection C, we propose the PSO-based desired speed optimization algorithm for performance improvement.

A. COLLISION PREDICTION

We consider the autonomy on the vehicle side as below. Each vehicle has to avoid collision with the forward vehicle and take measures to deal with emergencies. For the sake of simplicity, it is possible to set the normal acceleration and deceleration to a same-size $a_i^n > 0$. For example, $a_i^n = 2m/s^2$ indicates that the normal acceleration and deceleration of a vehicle are $2m/s^2$ and $-2m/s^2$. v_{max} is the road speed limit.

The safety distance sa_1 and sa_2 between V_i and its forward vehicle V_{i-1} in the same lane or the obstacle ahead are defined:

$$sa_1 = \frac{\max((v_i - v_{i-1}), 0)^2}{-2a_i^n} + \max\left(\frac{1}{2}(-a_i^n - a_{i-1})t_b^2, 0\right) + sa_0, \quad (14)$$

$$sa_2 = \frac{\max((v_i - v_{i-1}), 0)^2}{2a_{imin}} + \max\left(\frac{1}{2}(a_{imin} - a_{i-1})t_b^2, 0\right) + sa_0, \quad (15)$$

where t_b is response time for CAVs, and sa_0 is preset buffer distance for safety. Use X_i to represent the present position vector of vehicle V_i at the current time. If $\|X_i - X_{i+1}\| < sa_2$, it means V_i is too close to V_{i+1} and needs immediate emergency brake. Else if $\|X_i - X_{i+1}\| < sa_1$, it means V_i should take a safety brake to slow down properly and maintain distance from the forward vehicle. When the vehicle takes a brake result from the headway between V_i and the object in front of V_i is less than sa_1 , the pass permission should be withdrawn because this action is out of prediction by the EC server.

Assume that all vehicles have their own desired speed v_i^d . On the premise of $\|X_i - X_{i+1}\| > sa_1$, if the current speed $v_i < v_i^d$, V_i accelerates with the acceleration of a_i^n . Once the speed $v_i > v_i^d$, the vehicle decelerates to v_i^d with the acceleration of $-a_i^n$. It should be noted that the desired speed v_i^d vehicle V_i is set by no more than v_{i+1}^e of vehicle V_{i+1} if a safety brake occurs, to avoid unnecessary deceleration.

In practical situations, the internal mechanical control error of the vehicle and unpredictable external disturbance (e.g. wind and slope) could cause the deviation between practical status and prediction results, which can be fatal to vehicles. Therefore, PID control must be implemented in the vehicle side to reduce errors:

$$\begin{cases} u_i(t) = -K_o e - K_i \int e - K_d \frac{de}{dt}, \\ v_i^{act}(t) = K \cdot u_i(t), \end{cases} \quad (16)$$

where u is the input for the motor, K_o , K_i , K_d and K are PID controller coefficients and e is the error between actual velocity and expected velocity. The integral and derivative of e are applied to respectively offset historical error and correct the trend of error.

Although it is possible to give autonomy to vehicles, vehicles must follow the schedule of the EC server when considering the collision avoidance of the intersection.

The EC server gets all predictions $t_{i,k}^e$ and $t_{i,k}^l$ for all vehicles with formula (6), (9), (10), (13). In a large number of normal cases, $a_i = 0$, and in other cases, the acceleration a_i is fixed as described above. Therefore, $t_{i,k}^e$ and $t_{i,k}^l$ can be calculated using the formula of uniform acceleration motion. Algorithm 1 describes collision prediction algorithm of the proposed scheme. t_0 is the current time. I , A_n , X , v , v^d , S^e and S^l respectively represent matrix that are composed of $I_{i,j}$, $A_{i,n}$, X_i , v_i , v_i^d , $S_{i,j}^e$ and $S_{i,j}^l$. $S_{i,j}^e$ and $S_{i,j}^l$ are the distance from the current location of the vehicle V_i to the location entering and leaving the CBs in the $I_{i,j}$. With the above parameters, EC server predicts whether any two vehicles collide and expresses results with the collision prediction matrix P . If the resources of the two vehicles have overlapping parts, that is, the two vehicles do not satisfy the constraint (7), then the two vehicles may collide at the intersection. EC server always grasp the movement of all vehicles and runs prediction programs to ensure the stability and accuracy of prediction results. If there is a situation that the EC server does not anticipate, that is, vehicle V_i has slowed down significantly, then all $t_{i,k}^e$ and $t_{i,k}^l$ of V_i is set as infinity, because the state of the vehicle V_i can no longer be predicted when decelerating.

In algorithm 1, the collision prediction algorithm is presented. The first cycle (line 1 to line 21) is used to calculate the dwelling interval in the overlapping collision blocks of the two vehicles. The equation for calculating the entering and leaving time $t_{i,j}^e$ and $t_{i,j}^l$ is derived from the kinematic formula of uniform acceleration motion. We discuss the three situations including uniform motion (line 4 and line 5), uniform acceleration/deceleration (line 9 and line 14) motion and uniform motion (line 11 and line 16) after uniform acceleration/deceleration. The relationship between speed, acceleration/deceleration and target distance is analyzed under the limitation of maximum speed v_i^d . If the collisions of all the $T_{i,n}$ of all vehicles are calculated in turn according to the traversal rule, the computational complexity will be $O(M^2N^2)$. Using the previously defined set I , only the time t of entering and leaving the intersection is calculated, and the computational complexity is reduced to $O(N^2)$. If a predicted interval of V_i overlaps any occupied resource, for safety reasons, V_i should not be authorized to pass. Therefore, V_i and vehicles behind V_i are out of consideration in the next progress (line 22 to line 24). Then, predictions of the rest vehicles are recorded into the collision prediction matrix P for the next collision avoidance.

B. COLLISION AVOIDANCE

After discussions in subsection A, we obtained the collision prediction results of all vehicles traveling in the current state and stored them as a matrix P . In this subsection, we will describe how the EC server solves the collision avoidance of vehicles based on P , ensuring that vehicles pass through the intersection efficiently without collision. For vehicles approaching the intersection, a collection of vehicles should be found to pass the intersection and do not collide with each

Algorithm 1 Collision Prediction Algorithm for CAVs

Input: $t_0, I, A_n, X, v, v^d, S^e, S^l$

Output: P

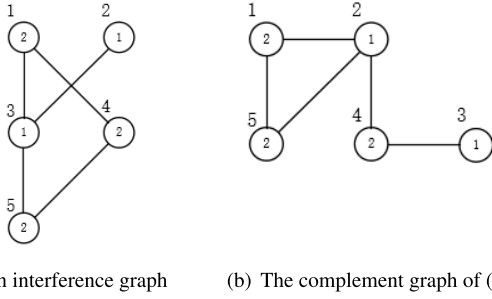
```

1: for each two vehicles  $i, j$  do
2:   if  $v_i = v_i^d$  then
3:      $t_{i,j}^e = t_0 + \frac{S_{i,j}^e}{v_i^d}$ 
4:      $t_{i,j}^l = t_0 + \frac{S_{i,j}^l}{v_i^d}$ 
5:   else if  $a_i > 0$  then
6:      $s_i^a = \frac{(v_i^d)^2 - (v_i)^2}{2a_i^n}$ 
7:     if  $s_i^a > S_{i,j}^e$  then
8:        $t_{i,j}^e = t_0 + \frac{|\sqrt{(v_i)^2 + 2a_i^n S_{i,j}^e} - v_i|}{a_i^n}$ 
9:     else
10:       $t_{i,j}^e = t_0 + \frac{|v_i^d - v_i|}{a_i^n} + \frac{S_{i,j}^e - s_i^a}{v_i^d}$ 
11:    end if
12:    if  $s_i^a > S_{i,j}^l$  then
13:       $t_{i,j}^l = t_0 + \frac{|\sqrt{(v_i)^2 + 2a_i^n S_{i,j}^l} - v_i|}{a_i^n}$ 
14:    else
15:       $t_{i,j}^l = t_0 + \frac{|v_i^d - v_i|}{a_i^n} + \frac{S_{i,j}^l - s_i^a}{v_i^d}$ 
16:    end if
17:  else
18:     $t_{i,j}^e \leftarrow -\infty$ 
19:     $t_{i,j}^l \leftarrow \infty$ 
20:  end if
21: end for
22: if Predicted interval of  $V_i$  overlaps occupied resource then
23:   Delete corresponding row and column of  $V_i$  and vehicles behind  $V_i$ 
24: end if
25: for each two vehicles  $i, j$  do
26:   if  $t_{i,j}^e > t_{j,i}^l$  &  $t_{j,i}^e > t_{i,j}^l$  then
27:      $P_{i,j} \leftarrow 1$ 
28:   else
29:      $P_{i,j} \leftarrow 0$ 
30:   end if
31: end for
32: return  $P$ 

```

other, for the purpose of minimizing the waiting time of all vehicles.

Considering the P as an adjacency matrix of a graph, we can draw the undirected interference graph $G = (V, E, \omega)$ for vehicles. If the two vehicles V_i and V_j are predicted to collide, then the edge $(i, j) \in E$. In this case, the EC server needs to find an independent set of vehicles that do not collide with each other to pass. If the unpermitted vehicle V_i is close to the intersection, and $s_i^l \leq \frac{v_i^2}{2a_i^n}$, V_i must slow down immediately with the acceleration of a_i^n to ensure it can stop outside the intersection.



(a) An interference graph (b) The complement graph of (a)

FIGURE 5. An example of an interference graph and its complement.

In order to ensure the fairness of the system, the vertex weights ω are defined as:

$$\omega_i(t) = q_i(t) + \xi \tau_i(t) + \eta v^d + \frac{\zeta}{s_i^1(t)}, \quad (17)$$

where ξ is the coefficient from current queue's waiting time $\tau_i(t)$ of V_i to the V_i 's current queue length $q_i(t)$. η and ζ respectively are the coefficient from v^d to $q_i(t)$ and $\frac{1}{s_i^1(t)}$ to $q_i(t)$. $\tau_i(t)$ begins counting when the vehicle stops in the queue and the calculation method of $q_i(t)$ has been given in [16]. With vertex weights increasing according to Eq. (17), vehicles with longer waiting times and vehicles in more congested directions are granted with higher priority. Especially, the fourth term in Eq. (17) indicates that the vehicle closer to the intersection has a higher priority. To better illustrate our approach, the following concepts are defined:

Definition 1 (The largest weighted independent set): Given a graph $G = (V, E, \omega)$, a vertex subset of a graph is called an independent set if any vertex in the subset are not adjacent in the graph. That is, if $U \subseteq V$, and for any two vertexes $u, v \in U$ has $(u, v) \notin E$, then U is an independent set of G , where an unordered pair (u, v) represents an edge between vertex u and v . The maximum weighted independent set D_I is an independent set with the maximum sum of weights of all vertexes, which can be written as follows:

$$D_I = \arg \max_D \sum_{i \in D \subseteq V} \omega_i(t), \quad (18)$$

where D is an independent set of graph G . In our model, the optimal solution is to find ID_I in each processing interval. Since the maximum weighted independent set is a set of isolated points, it is difficult to be solved by algorithms. We take the complement graph for the interference graph, and analyze the complement graph of the interference graph. Let G 's complement graph be $\tilde{G}$, $\tilde{G} = (V, E', \omega)$, where $E' = \{(u, v) | \forall (u, v) \notin E\}$.

Fig. 5(a) describes an example of an interference graph and its complement. The number in the vertex represents its vertex weight, and the number near the vertex is its serial number. It is not easy to find the maximum weighted independent set of the graph in Fig. 5(a). When taking the complement for the interference graph, the set of vertexes with no edges connected to each other in the original graph has converted to

the set where all vertexes are connected to other vertexes. The objective can be transformed to get the weighted maximum clique. In the case which is depicted in Fig. 5, we can easily get that the solution set is $\{1, 2, 5\}$ as its sum weight of vertexes equals to 5.

Definition 2 (The weighted maximum clique): Given a graph $G = (V, E, \omega)$, a subgraph of a graph is called a clique, if any vertex in the subgraph is adjacent with the others. That is, if $U \subseteq V$, and for any two vertexes $u, v \in U$ has $(u, v) \in E$, then $G' = (U, E, \omega)$ is a clique of G . The weighted maximum clique of G refers to the clique of G where it maximizes the sum of all vertex weights.

Let C be a clique of $\tilde{G}$. With the above analysis, the objective function is given as follows:

$$f(C) = \sum_{i \in C \subseteq \tilde{G}} \omega_i(t). \quad (19)$$

A common method for solving the problem of the weighted maximum clique (WMC) problem is the Bron-Kerbosch algorithm, which is used to find the optimal solution. However, its NP-hardness will bring a heavy burden to the computing unit of the EC server. Besides, because of the time-critical feature of an AIC system, it will cause a great impact on traffic safety and transportation efficiency if the result cannot be calculated within a limited time. Moreover, although AIC can be modeled as a WMC problem, there are still differences between them as special constraints. That is, a vehicle is able to be permitted only if all its previous vehicles on the same lane is permitted to pass through the intersection, or it will cause extra useless behavior for vehicles and extra consumption of computational resources. It is called as the sequential-passing limitation. Hence, the existing algorithms on the WMC problem cannot be directly utilized for our problem, and efficient heuristic algorithms are required.

We propose an AIC algorithm using a tabu search algorithm, which is based on solving the WMC problem. First, we set a lane queue $lq^1 - lq^8$ for each lane. If a vehicle V_i enters the management region of an intersection at lane l^k , then V_i will enter the queue lq^k in the EC server. All queues follow the first-in-first-out rule. The vehicle V_i dequeues only when the vehicle V_i obtains permission to pass the intersection. Denote lq_n^k as the n th element in lq^k , and V^C is the vertex set of C . Then the sequential-passing limitation can be mathematically written as:

$$V^C \models \forall i \in V^C \wedge i = lq_n^k \wedge n \geq 2 : lq_{n-1}^k \in V^C. \quad (20)$$

For the complementary graph $\tilde{G}$ of the interference graph G , we divide its vertex into four parts defined as follows:

- **CS.** CS is the set of vertexes of the currently chosen clique, which represents a solution to IM problem.
- **PA.** A vertex i in the set PA satisfies that it is connected to all of the vertexes in CS. PA represents a set of vertexes that is currently candidate to be added to CS.
- **OM.** A vertex i in the set OM satisfies that it is only boundless to one vertex in CS. PA represents a set of

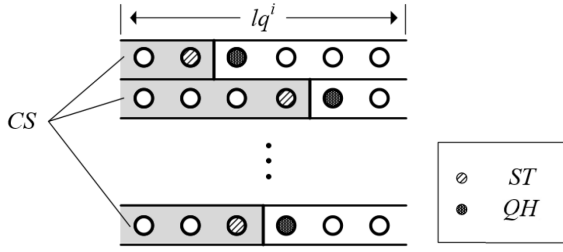


FIGURE 6. The relationship between QH , ST , lq and CS .

vertexes that is currently candidate to be swapped with the vertex unconnected to i in CS .

- **RS.** RS is the rest set of vertexes excluded from the set CS , PA and OM in graph $\tilde{G}$.

Define two sets QH and ST . The relationship between QH , ST , lq and CS is visually displayed in Fig. 6. QH is the set of queue heads of $(lq^1 - CS)$ to $(lq^8 - CS)$, and ST is the set that contains the top of stacks differentiate by lane which stores the vertex in CS . All operations will be done between the vertexes on either side of the line which separating ST and QH to satisfy the constraint (8). Then three kinds of neighborhood operations are defined:

- **ADD**(i, S). Select a vertex i randomly from the set S to join the CS .
- **SWAP**(i, S). Select a vertex i randomly from the set S , and swap out the vertex which i is boundless from CS .
- **DROP**(i, S). Select a vertex i randomly to remove from the set S .

Note that these three operations are not absolutely optimal. Sometimes SWAP will generate a larger objective function increment than ADD, whereas DROP represents the acceptance of inferior solutions in tabu search. Queue-based tabu search algorithm for finding maximal clique MQ is shown in Algorithm 2. The setting rules for the tabu list for the three kinds of operations are:

- 1) For the operation ADD, since the objective function is going to be maximized, the probability of the point at which the solution enters is removed is small, and the length of the tabu should be set smaller, which is set to 4 here.
- 2) For the operation SWAP, set the tabu for swapped vertexes in CS , the tabu length is the basic tabu length (which is set to 6 here) plus the random number from 0 to $|OM|$.
- 3) For the operation DROP, set a tabu for each removed point with the basic tabu length.

When a vertex i is in the tabu list of ADD, it means that the vertex cannot be performed an add operation in the next l operations (assume current tabu length is l). The tabu lists for SWAP and DROP are the same. However, there is an exception for a tabued operation if the global optimal solution obtained can be updated when performing this operation. In this case, the optimal solution is updated to this solution.

Algorithm 2 Queue-Based Collision Avoidance Algorithm

Input: $\tilde{G}, lq$

Output: MQ, r

```

1:  $CS^m \leftarrow \emptyset, r^m \leftarrow 0$ 
2: while  $loop < \text{maximum number of repeating calculation}$ 
    $\Delta$  do
3:   add the vertex with largest weight into  $CS$ 
4:   while  $PA \neq \emptyset$  do
5:     get current  $PA$ 
6:      $ADD(i, PA)$ 
7:     where  $i$  is fixed by  $i = \arg \max_{i \in PA} \mu_i$ 
8:      $r = \text{calR}(\emptyset)$ 
9:     if  $r > r^m$  then
10:       $CS^m \leftarrow CS, r^m \leftarrow r$ 
11:    end if
12:  end while
13:   $Ta \leftarrow \emptyset, Ts \leftarrow \emptyset, Td \leftarrow \emptyset,$ 
14:  while  $iter < \text{maximum number of iteration } \Theta$  do
15:    get current  $PA, OM$  and  $QH$ 
16:    if  $PA \neq \emptyset$  and  $OM \neq \emptyset$  then
17:       $Rp \leftarrow \text{getR}(PA), Ro \leftarrow \text{getR}(OM)$ 
18:      if  $Rp > Ro$  then
19:         $ADD(i, (PA \cap QH) - Ta)$ 
20:      else
21:         $SWAP(i, (OM \cap QH) - Ts)$ 
22:      end if
23:    else if  $PA \neq \emptyset$  and  $OM = \emptyset$  then
24:       $ADD(i, (PA \cap QH) - Ta)$ 
25:    else if  $PA = \emptyset$  and  $OM \neq \emptyset$  then
26:       $SWAP(i, (OM \cap QH) - Ts)$ 
27:    else
28:       $DROP(i, CS - Td)$ 
29:    end if
30:     $\text{calT}(Ta, Ts, Td)$ 
31:     $r = \text{calR}(\emptyset)$ 
32:    if  $r > r^m$  then
33:       $CS^m \leftarrow CS, r^m \leftarrow r$ 
34:    end if
35:    if  $r^m$  does not change for 15 iterations then
36:      break
37:    end if
38:  end while
39: end while
40:  $r \leftarrow r^m$ 
41: return  $MQ, r$ 

```

Algorithm 2 describes a procedure that the EC server finds a solution. The algorithm uses the greedy algorithm to construct the initial solution so that the algorithm obtains a better initial state from the beginning. To avoid its falling into the local optimal solution, the EC server adopts several independent repeated calculation and take the best solution from these results. In each iteration $iter$ in each calculation (line 14 to line 38), the operation that maximizes the reward increment

in current available operation is executed. For the operation ADD and SWAP, the intersection of PA and QH or OM and QH is taken to ensure that the solution meets the constraint (7). $\text{calR}(\cdot)$ is used to calculate the value of function $f(CS)$ after an operation. Ta , Ts , Td respectively indicate the tabu set for the operation ADD, SWAP and DROP. The function $\text{calT}(T)$ updates the tabu set T . The vertex operated in a loop is added to the tabu list of the corresponding operation, and then tabu length is updated following the rules given above. For vertexes that are not operated in this loop, all tabu lengths decrement by 1. When the tabu length of a vertex is reduced to 0, it is removed from the tabu set.

Denote the maximum number of repeating calculations and the maximum number of iteration in a calculation as Λ and Θ , respectively. Λ and Θ can be adjusted according to the computing capacity of the EC server. If the value of reward does not change for many iterations, it deems that the algorithm convergences. Therefore, the loop can be broken early to save computational time. Multiple repeats of the solution are taken (line 2) to prevent the result from falling into the local optimum. Generally, $\mathcal{M} > 5$. Therefore, algorithm 2 achieves low complexity with linear computational complexity of $O(\Theta)$.

C. PARTICLE SWARM OPTIMIZATION

In this subsection, a PSO-based cooperative optimization is proposed to facilitate mutual cooperation among vehicles. By coordinating the desired speed of all vehicles, it further promotes the space utilization.

Particle swarm optimization (PSO) is a global random search algorithm inspired by migration and clustering behavior of foraging birds within a flock [37], [38]. The PSO has shown a successful performance in various modern engineering problems [39]. In PSO, each potential solution is regarded as a position of a particle and all particles are a swarm. Each particle searches in the solution space and its position is updated each iteration g by means of:

$$x_i^{g+1} = x_i^g + v_i^{g+1}, \quad (21)$$

where v_i^{g+1} is the velocity of particle i , which is given by

$$v_i^{g+1} = \omega \cdot v_i^g + C_1 \cdot R_1(\rho_i^g - x_i^g) + C_2 \cdot R_2(b^g - x_i^g). \quad (22)$$

In this formula, ρ_i^g is the best solution that the particle i has seen discovered, and b^g is the global best solution in the history of the swarm. ω is the inertia weight (it can be adjusted to balance the global optimization and local optimization performance). C_1 and C_2 are called acceleration constants, where the former is the individual learning factor for each particle, while the latter is the social learning factor. Suganthan's experiments [40] show that a preferable solution is obtained when C_1 and C_2 are constant (generally $C_1 = C_2 = 2$). Finally, R_1 and R_2 are random numbers in the interval $[0,1]$, which is resampled for each component of the velocity vector, for each particle and iteration.

Algorithm 3 describes the pseudocode for performance optimization using PSO. We input the desired speed v^d of

the scheduling vehicles as the position of the particles. In the beginning, the swarm is initialized, which contains the particles' velocity and position. ρ_i is set to the current position of each particle, and the global best position is the best particle in the swarm. All particles are searching for the maximum reward in the solution space. For each iteration, particles search for reward and update their velocity and position (line 5 and line 6). Then, the personal best position of particles and the global best position is updated in each iteration.

The penalty function method is implemented to avoid the unreasonably of solutions. The velocity and position of each particle should satisfy that:

$$-v_{max} \leq v_i^g \leq v_{max}, \quad (23)$$

$$v_{min} \leq x_i^g \leq v_i^d. \quad (24)$$

If v_i^g or x_i^g of a particle exceeds the limitation, it will be set to the boundary value. Besides, a vehicle cannot enter the intersection before the preceding vehicle has not entirely entered the intersection, that is,

$$t_{i,first}^e - t_{i-1,first}^l > 0. \quad (25)$$

Then, the penalty function can be expressed as:

$$penalty = \begin{cases} 0, & t^{ur} > 0, \\ \delta \cdot t^{ur}, & t^{ur} \leq 0, \end{cases} \quad (26)$$

where

$$t^{ur} = t_{i,first}^e - t_{i-1,first}^l, \quad (27)$$

δ is a large positive number to make the solution after the penalty is impossible to become a good solution. Consequently, the reward r can be given by:

$$r = r' + penalty, \quad (28)$$

where r' is the maximum value of function $f(C)$ calculated by algorithm 2.

Algorithm 3 PSO-Based Desired Speed Optimization

Input: v^d

Output: v^d

- 1: initializeSwarm()
 - 2: $b_i = \text{computeGlobalBest}()$
 - 3: **while** $g < p$ **do**
 - 4: **for** each particle x_i^g **do**
 - 5: $v_i^{g+1} = \text{updateVelocity}(\rho_i^g, b^g)$
 - 6: $x_i^{g+1} = \text{updatePosition}(x_i^g, v_i^{g+1})$
 - 7: $r_i^{g+1} = \text{computeSolution}(x_i^{g+1})$
 - 8: $\rho_i^{g+1} = \text{update}(\rho_i^g)$
 - 9: **end for**
 - 10: $b^{g+1} = \text{computeGlobalBest}(b^g, r_i^{g+1})$
 - 11: **end while**
 - 12: $v^d = b^p$
 - 13: **return** v^d
-

Algorithm 3 continuously seeks the coordination of the desired speed of vehicles to acquire a larger reward. For unpermitted vehicles in the final solution, their new v^d can make a Pareto improvement by increasing them to the value before scheduled by Algorithm 3, without any influence in the permitted vehicles. The EC server does not need to change the desired speed of unpermitted vehicles. After the new v^d is calculated, the EC server updates the resource occupancy of the intersection, and then sends the adjusted v^d and permission to corresponding vehicles according to the solution calculated by Algorithm 2 in subsection B. Since all iterations in the scheme can be set in advance, the whole solution can be obtained at a predictable time. Denote the iteration number of PSO and the number of particles as p and n , respectively. The complexity of the whole solution is $O(p \cdot \max(N^2, \Theta))$ for parallel computing and $O(n \cdot p \cdot \max(N^2, \Theta))$ for serial computing, which is able to make real-time scheduling even in time.

V. PERFORMANCE EVALUATION

A. THEORETICAL UPPER LIMIT

Consider an extreme situation that all vehicles do not interfere with each other when passing through the intersection, we have the optimal indicator of intersection management problems. Although it is absolutely impossible to achieve in the actual situation, it provides an upper bound for the optimization goal. We consider the Greenshields model [41], [42] for analyzing the theoretical upper limit of vehicles' average speed through the travel.

Assume that vehicle flow is Q in all lanes. Suppose k^j is the jam density and k is the current density, and let v^f represents the free-flow speed. The current vehicle speed is given by:

$$v = v^f \left(1 - \frac{k}{k^j}\right). \quad (29)$$

Take expectations on both sides, we have

$$E[v] = v^f \left(1 - \frac{E[k]}{k^j}\right). \quad (30)$$

Take expectations the equation $v = \frac{Q}{k}$, the relationship between $E[v]$ and $E[k]$ is obtained as follows:

$$E[v] = \frac{E[Q]}{E[k]}. \quad (31)$$

We can derive the expected speed of the traffic flow $E[v]$, i.e., the upper limit of vehicles' average speed, from the formula (30) and (31):

$$E[v] = \frac{v^f}{2} + \sqrt{\left(\frac{v^f}{2}\right)^2 - \frac{qv^f}{k^j}}, \quad (32)$$

where q is the average traffic flow in a lane.

B. EXPERIMENT SETUP

The simulation model is developed by Matlab for traffic simulation and algorithm implementation, where the intersection

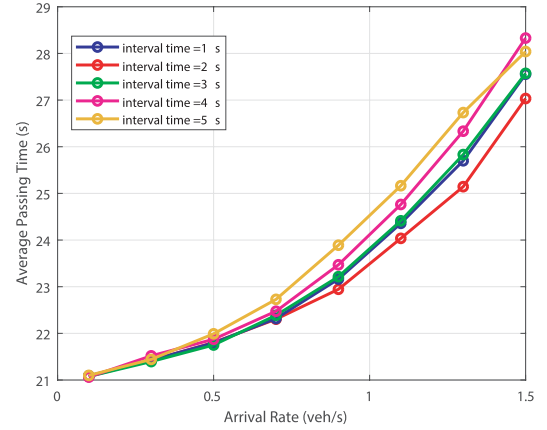


FIGURE 7. Average passing time under different traffic loads with different schedule interval time.

model is designed under the model of Fig. 4. In our simulation, we assume that lane-changing is not allowed, there is no obstacle to avoid and all vehicles have the same length and acceleration a_i^n and a_{imax} setups. In order to be more realistic, the vehicles' velocity has unpredictable jitter during the travel. Vehicles are evenly generated from each lane in all directions. The initial velocity of new arrival vehicles is generated, subject to the normal distribution with a mean of 10 km/h and a variance of 0.25. The desired speed of vehicles on the road will approximate equation (32) according to the Greenshield's model. In this model, we have $v^f = 10$ m/s and $k^j = 150$ veh/km. The normal acceleration and deceleration a_i^n is set to 2 m/s² and the maximum deceleration is set to -5 m/s².

C. PARAMETERS DETERMINATION

To determine the optimal parameters of our algorithm, i.e., the schedule interval time, We performed the following simulation. The arrival pattern of vehicles follows the Poisson process with an arrival rate of λ of the system. We test the performance of the algorithm under different traffic densities and the vehicle arrival rate λ is varying from 0.5 to 1.5, which corresponds to the traffic throughput of is 360 to 5400 veh/h for the system.

We evaluate the average pass time (APT) through the intersection under different arrival rates, which represents the average of the time interval between vehicles entering the management region and leaving the departure region. We take the simulation time as 3600 s for each arrival rate.

Fig. 7 shows the APT under different traffic load with different schedule interval time. A lower value of APT indicates that the mode of interval time achieves better scheduling in terms of traffic efficiency. As shown in Fig. 7, APTs are not sensitive to the changes in interval time when λ is below 0.5 veh/s. Because in relatively low traffic density, the interference graph is not complicated, so the EC server makes similar decisions under different interval times. As the arrival rate increases, it requires a more complex solution. However, long interval time brings a higher decision delay

for vehicles. When interval time equals to 2 s, APT has the smallest increment when λ increases, then the interval time is set to 2 s, consequently.

D. BENCHMARK METHOD

For performance comparison, we implement the other two schemes:

a) Traffic Light (TL) is a conventional traffic control strategy where a signal light is used to control the permission of vehicles in each direction passing through. In our simulation, the time interval between the traffic light switching a phase to another is set to 30 s.

b) FEFS [23] is a heuristic approach, which aims at resolving conflicts efficiently and timely by scheduling the passing time of conflict points with the overlapped trajectory.

c) QEOIC [26] is a velocity assignment algorithm. The algorithm pays attention to the passengers' quality-of-experience (QoE) and obtains more by minimizing the value of the acceleration adjustment.

d) SREIC [31] is a reservation-based AIC algorithm. By allowing vehicles to travel in a speed range, the algorithm resists external disturbances.

To evaluate the performance of our system, we assume that vehicles arrive according to a Poisson distribution. Note that due to the limitation of safety headway, the actual arrival rate generated may not exactly as the specified value when the arrival rate is high. The values of parameters used in the simulation are summarized in Table 3.

E. METRICS

We define the following metrics to measure the performance of the proposed scheme in various aspects:

a) Average waiting time (AWT): it is defined as the average waiting time that a vehicle waits for passing in front of the intersection. Short waiting time indicates vehicles will get better quality of experience (QoE) during the travel.

b) Average speed ($\bar{v}$): the average speed of passing through the intersection for all vehicles. Denote s_i as the distance that vehicle i travels through the area which the EC controls, and t_i^e, t_i^l represent the time when vehicle i enters and leaves the area, respectively. $\bar{v}$ is given by:

$$\bar{v} = \frac{1}{N} \sum_{i=1}^N \frac{s_i}{t_i^l - t_i^e}. \quad (33)$$

c) Standard deviation of speed ($S_{\bar{v}}$): $S_{\bar{v}}$ reflects the overall deviation of vehicles' speed from $\bar{v}$. $S_{\bar{v}}$ is computed by:

$$S_{\bar{v}} = \sqrt{\frac{1}{N} \sum_{i=1}^N (v_i - \bar{v})^2}. \quad (34)$$

A lower value of $S_{\bar{v}}$ indicates that the system has a more balanced strategy for all vehicles, and the travel time can be predicted more accurately.

d) Throughput: throughput is defined as the number of vehicles passing through the intersection per hour.

TABLE 3. Parameter values.

Parameter	Value
Maximum Speed	10 m/s ²
Normal Acceleration	2 m/s ²
Maximum Acceleration	5 m/s ²
Management Region	150 m
Departure Region	50 m
Arrival Rate	0.1 to 1.5 veh/s
Simulation Time	3600 s

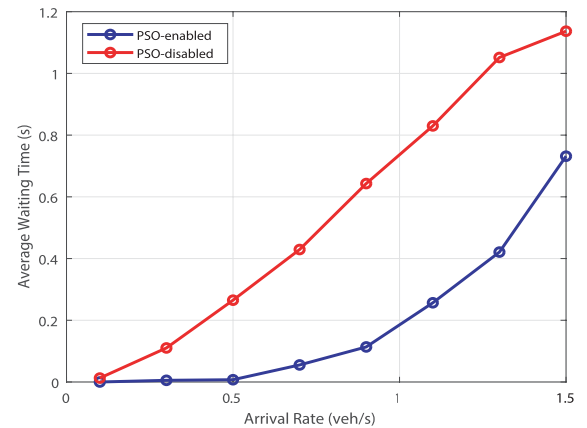


FIGURE 8. AWT comparison between PSO-enabled and PSO-disabled algorithm.

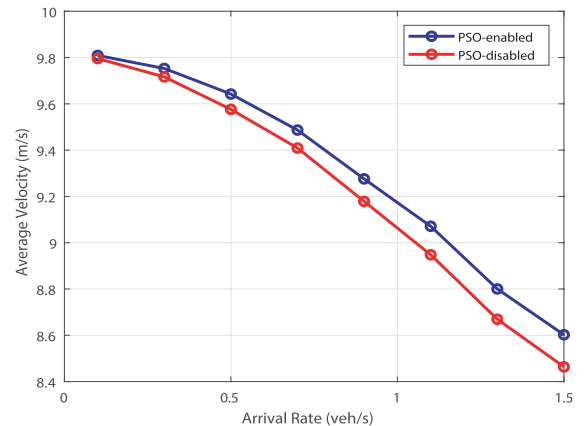


FIGURE 9. Average speed comparison between PSO-enabled and PSO-disabled algorithm.

Throughput reflects whether the algorithm is good at enhancing traffic efficiency.

F. SIMULATION RESULTS

We conduct a comparison to demonstrate the effect of PSO in our scheme. Fig. 8 and Fig. 9 respectively depict the AWT and $\bar{v}$ performance between PSO-enabled and PSO-disabled strategy in the proposed scheme. In comparison with $\bar{v}$ of two strategies, they are almost the same when the traffic volume is small. As the vehicle arrival rate increases, the performance of PSO-enabled strategy exceeds the PSO-disabled strategy slightly. Meanwhile, the AWT has been greatly reduced after applying PSO. The mechanism of adjusting v^d decreases by

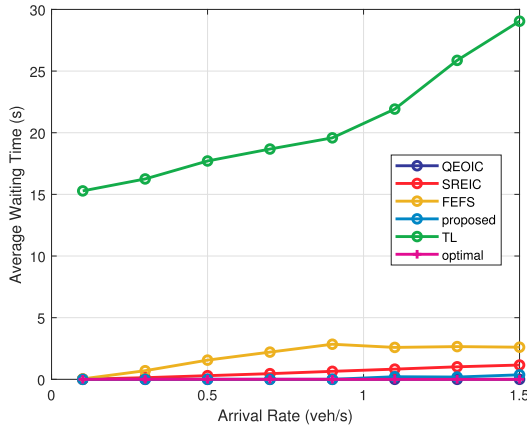


FIGURE 10. Average waiting time under different arrival rates.

more than 30% AWT. Especially, when the vehicle arrival rate is less than 1 veh/s, vehicles has almost no waiting time (< 0.1 s), which will substantially improve the passenger's driving experience.

Fig. 10 illustrates the AWT that the proposed scheme compares with other algorithms. AIC schemes achieve significantly lower AWT than TL. TL has the largest AWT due to its fixed phase duration of red light, regardless of the realistic traffic condition. When comparing between reservation-based algorithms, the proposed scheme outperforms FEFS and SREIC. The reason is that in these two algorithms, the passage of vehicles is scheduled by the arrival order. When the conflict situations between vehicles become complex, FEFS and SREIC fail to find proper speeds for every vehicle to pass, causing their waiting. AWT of QOEIC outperforms the proposed scheme and reaches to the optimal performance because QOEIC reduces waiting with real-time adjustments. Although it does not reach to the optimal, the proposed scheme also achieves a near-optimal performance (close to 0) by scheduling vehicles from a system perspective, optimizing the resources. In this way, the utilization of the intersection area is improved, and AWT is reduced.

Fig. 11 shows the average speed $\bar{v}$ of different schemes as a function of arrival rate. The simulation result indicates that it shows different characteristics between AIC schemes and TL in response to increased vehicle arrivals. When comparing AIC schemes, the $\bar{v}$ in FEFS decreases rapidly when the arrival rate increases while the proposed scheme, QOEIC and SREIC has advantages in keeping the vehicles' speed, relatively. Among AIC schemes, the proposed scheme has better performance in terms of $\bar{v}$ with the average speed always close to the optimal. In TL, its $\bar{v}$ has the smallest variation with the arrival rate. However, it displays the worst performance resulting from its long waiting time, as demonstrated in Fig. 10. The advantage of the proposed scheme is mainly caused by two reasons. Firstly, the proposed scheme does not have the limitations of the first-come-first-served strategy, which enables it to make better solutions in a larger solution space. Secondly, the PSO-based optimization designed in

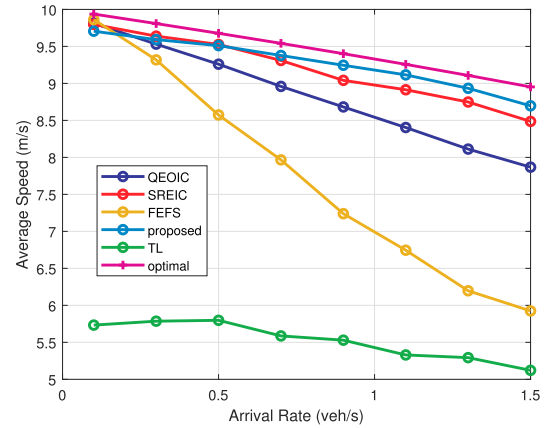


FIGURE 11. Average speed under different arrival rates.

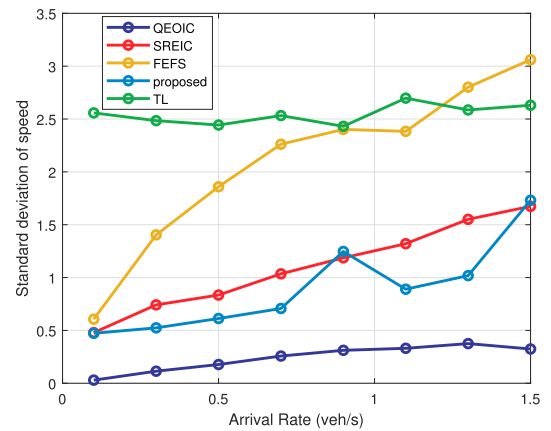


FIGURE 12. Standard deviation of speed under different arrival rates.

Section IV-C is used to coordinate the actions of all vehicles, achieving higher space utilization.

In terms of the standard deviation of speed, the simulation result is depicted in Fig. 12. As discussed above, the value of $S_{\bar{v}}$ indicates the fairness of the system. The proposed scheme achieves more stable $S_{\bar{v}}$ than FEFS and TL. Meanwhile, the proposed scheme achieves comparable performance with SREIC. The advantage of the proposed scheme compared with SREIC, FEFS and TL is caused by the following two reasons. On the one hand, the non-first-come-first-served that the proposed scheme adopted reduces the waiting, which is the situation that easy to cause an individual average speed deviating from the overall average speed. On the other hand, to guarantee the system fairness, we the proposed scheme take waiting time and queue length into consideration. Vehicles with long waiting times and those in the lanes with long queues have a higher priority to pass. Therefore, the $S_{\bar{v}}$ of the proposed scheme maintains a lower level. Without any doubt, QOEIC outperforms other schemes in terms of $S_{\bar{v}}$. As QOEIC takes minimum speed differences as one of its optimization goals to improve passengers' QoE, vehicles are scheduled to be approximate to an desired speed, which achieves excellent system fairness of QOEIC.

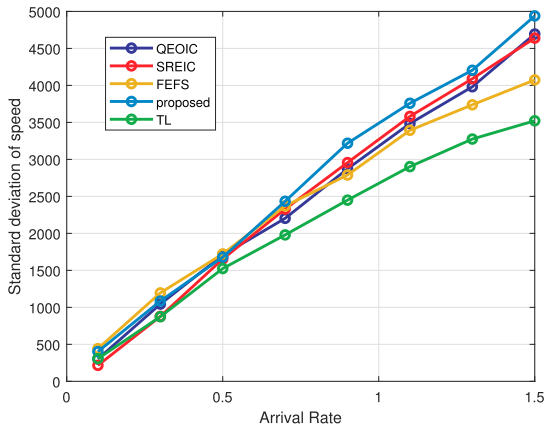


FIGURE 13. Throughput under different arrival rates.

However, QOEIC sacrifices transportation efficiency. According to Fig. 11 and Fig. 12, we can conclude that the proposed scheme achieves a decent balance between traffic efficiency and system fairness.

Throughput is an important consideration indicating the transportation efficiency for AIC systems. A linearly increasing throughput with arrival rate indicates that the intersection is smooth in traffic. From the figure, we can observe that the growth rate of throughput becomes smaller, the intersection starts to suffer from congestion. As depicted in Fig. 13, the proposed scheme outperforms others and maintains near-linear growth as the arrival rate increases, which suggests that the proposed scheme maintains high traffic efficiency in the face of a high workload. When the traffic flow is sparse, FEFS outperforms the proposed scheme slightly. However, it starts to suffer from congestion after the arrival rate of 0.9 veh/s. The proposed scheme is in better performance of throughput, which suggests that it has decent performance in terms of traffic efficiency.

G. EFFECTS OF PACKET LOSS

We conduct a simulation for the effects of packet loss to evaluate the robustness of our proposed scheme. Note that the compared algorithms are not comparable because they do not consider the packet loss issue. We assume that the EC server can get a vehicle's all information only from the vehicle itself. In our proposed scheme, if a vehicle does not receive a downlink packet containing instructions in time, the vehicle continues to move with its own decision. And if an uplink packet sent by a vehicle is lost, the EC server calculates the current status of this vehicle by its previous kinematic status and does not allow it to enter the intersection area until the EC server receives new packets from this vehicle.

As shown in Fig. 14 and Fig. 15, the performance of S_v and throughput decrease with the rise of packet loss rate. Compared with the performance in the ideal situation, the S_v increases within an acceptable increment when the percentage of packet loss is below 10%. However, in 20% packet loss case, the S_v goes up sharply because some vehicles

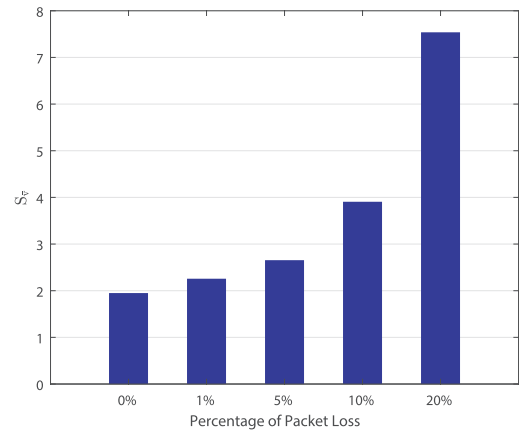


FIGURE 14. Standard deviation under different packet loss rates.

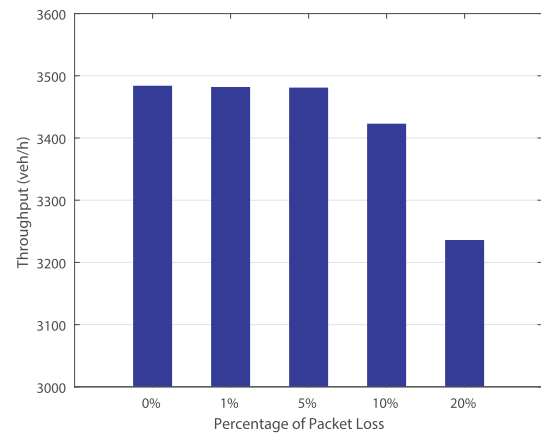


FIGURE 15. Throughput under different packet loss rates.

always can not receive instructions. Compared with throughput, although throughput decreases as the packet loss rate increases, the throughput stays basically stable in low packet loss rate cases (1%, 5% and 10%), and still maintains satisfying performance (7.1% decline compared with ideal performance) in the environment of 20% packet loss rate. From the analyses in the performance of S_v and throughput above, we can conclude that the proposed scheme performs good robustness in the environment of packet loss. In particular, the proposed scheme keeps its efficiency even in a poor network quality (20% packet loss).

VI. CONCLUSION AND FUTURE WORK

With the development of telecommunication technology, edge-assisted CAVs are becoming a great solution for high-level autonomous driving. In this paper, we have proposed a robust and efficient solution for edge-assisted AIC systems. Specifically, we first define the management region and divide the intersection into collision blocks. By dividing the resources in space and time dimension for model simplification, we analyze the intersection control problem. Then, we formulate the model into a weighted maximal clique problem. A stack-based tabu search algorithm is implemented to solve this NP-hard problem. Furthermore, we introduce a

PSO algorithm to dynamically adjust the vehicles' desired speed. By implementing a PSO algorithm, the efficiency further optimized. Finally, the work procedure on the EC server-side is elaborated. Simulation results illustrate that our proposed AIC solution achieves decent throughput while maintaining system fairness and robustness.

In the future, we plan to extend our work from single intersection management to combination optimization among multiple intersections. Moreover, a realistic testbed is expected to be built to verify our current solution in a realistic system.

REFERENCES

- [1] Q. Zhu, Q. Mao, L. Chen, M. Li, and Q. Li, "VeloRegistration based intersection detection for autonomous driving in challenging urban scenarios," in *Proc. 15th Int. IEEE Conf. Intell. Transp. Syst.*, Sep. 2012, pp. 1191–1196.
- [2] Volvo Trucks, "European accident research and safety report," Volvo Trucks, Gothenburg, Sweden, Tech. Rep., 2013. [Online]. Available: <https://www.volvogroup.pe/content/dam/volvo/volvo-group/markets/global/en-en/about-us/traffic-safety/ART-report-2013.pdf>
- [3] M. C. Simon, T. Hermitte, and Y. Page, "Intersection road accident causation: A European view," in *Proc. 21st Int. Tech. Conf. Enhanced Saf. Veh.*, Jun. 2009, pp. 1–10.
- [4] G. Yannis, P. Evgenikos, P. Papantoniou, J. Broughton, J. Knowles, C. Brandstaetter, N. Candappa, M. Christoph, K. van Duijvenvoorde, M. Vis, and J. F. Pace, *Traffic Safety Basic Facts 2012: Single Vehicle Accidents*. Brussels, Belgium: DaCoTA EU Project, 2013. [Online]. Available: <http://www.dacota-project.eu/Deliverables/BFS%20ASR%202012/BFS2012-Dacota-Ntua-single%20vehicle%20accident.pdf>
- [5] D. Schrank, B. Eisele, T. Lomax, and J. Bak, *2015 Urban Mobility Scorecard*. College Station, TX, USA: Texas A&M Transp. Inst., 2015.
- [6] Y. Shi, Y. Pan, Z. Zhang, Y. Li, and Y. Xiao, "A 5G-V2X based collaborative motion planning for autonomous industrial vehicles at road intersections," in *Proc. IEEE Int. Conf. Syst., Man, Cybern. (SMC)*, Oct. 2018, pp. 3744–3748.
- [7] C. Wu, Z. Liu, D. Zhang, T. Yoshinaga, and Y. Ji, "Spatial intelligence toward trustworthy vehicular IoT," *IEEE Commun. Mag.*, vol. 56, no. 10, pp. 22–27, Oct. 2018.
- [8] S. Guleng, C. Wu, Z. Liu, and X. Chen, "Edge-based V2X communications with big data intelligence," *IEEE Access*, vol. 8, pp. 8603–8613, 2020.
- [9] J. Feng, Z. Liu, C. Wu, and Y. Ji, "AVE: Autonomous vehicular edge computing framework with ACO-based scheduling," *IEEE Trans. Veh. Technol.*, vol. 66, no. 12, pp. 10660–10675, Dec. 2017.
- [10] Y. Mao, C. You, J. Zhang, K. Huang, and K. B. Letaief, "A survey on mobile edge computing: The communication perspective," *IEEE Commun. Surveys Tuts.*, vol. 19, no. 4, pp. 2322–2358, 4th Quart., 2017.
- [11] J. Feng, Z. Liu, C. Wu, and Y. Ji, "Mobile edge computing for the Internet of vehicles: Offloading framework and job scheduling," *IEEE Veh. Technol. Mag.*, vol. 14, no. 1, pp. 28–36, Mar. 2019.
- [12] V. S. Varanasi and S. Chilukuri, "Adaptive differentiated edge caching with machine learning for V2X communication," in *Proc. 11th Int. Conf. Commun. Syst. Netw. (COMSNETS)*, Jan. 2019, pp. 481–484.
- [13] A. Moubayed, A. Shami, P. Heidari, A. Larabi, and R. Brunner, "Edge-enabled V2X service placement for intelligent transportation systems," *IEEE Trans. Mobile Comput.*, early access, Oct. 10, 2020, doi: [10.1109/TMC.2020.2965929](https://doi.org/10.1109/TMC.2020.2965929).
- [14] K. Dresner and P. Stone, "A multiagent approach to autonomous intersection management," *J. Artif. Intell. Res.*, vol. 31, pp. 591–656, Mar. 2008.
- [15] J. Garcia-Nieto, A. C. Olivera, and E. Alba, "Optimal cycle program of traffic lights with particle swarm optimization," *IEEE Trans. Evol. Comput.*, vol. 17, no. 6, pp. 823–839, Dec. 2013.
- [16] W. Liu, G. Qin, Y. He, and F. Jiang, "Distributed cooperative reinforcement learning-based traffic signal control that integrates V2X Networks' dynamic clustering," *IEEE Trans. Veh. Technol.*, vol. 66, no. 10, pp. 8667–8681, Oct. 2017.
- [17] M. A. Khamis and W. Gomaa, "Adaptive multi-objective reinforcement learning with hybrid exploration for traffic signal control based on cooperative multi-agent framework," *Eng. Appl. Artif. Intell.*, vol. 29, pp. 134–151, Mar. 2014.
- [18] R. Hoar, J. Penner, and C. Jacob, "Evolutionary swarm traffic: If ant roads had traffic lights," in *Proc. Congr. Evol. Comput. CEC*, May 2002, pp. 1910–1915.
- [19] M. Khayatian, M. Mehrabian, and A. Shrivastava, "RIM: Robust intersection management for connected autonomous vehicles," in *Proc. IEEE Real-Time Syst. Symp. (RTSS)*, Dec. 2018, pp. 35–44.
- [20] S. Noh, "Decision-making framework for autonomous driving at road intersections: Safeguarding against collision, overly conservative behavior, and violation vehicles," *IEEE Trans. Ind. Electron.*, vol. 66, no. 4, pp. 3275–3286, Apr. 2019.
- [21] C.-M. Huang, M.-S. Chiang, D.-T. Dao, W.-L. Su, S. Xu, and H. Zhou, "V2 V data offloading for cellular network based on the software defined network (SDN) inside mobile edge computing (MEC) architecture," *IEEE Access*, vol. 6, pp. 17741–17755, 2018.
- [22] L. Li and F.-Y. Wang, "Cooperative driving at blind crossings using intervehicle communication," *IEEE Trans. Veh. Technol.*, vol. 55, no. 6, pp. 1712–1724, Nov. 2006.
- [23] A. P. Chouhan and G. Banda, "Autonomous intersection management: A heuristic approach," *IEEE Access*, vol. 6, pp. 53287–53295, 2018.
- [24] Z. He, L. Zheng, L. Lu, and W. Guan, "Erasing lane changes from roads: A design of future road intersections," *IEEE Trans. Intell. Veh.*, vol. 3, no. 2, pp. 173–184, Jun. 2018.
- [25] R. Krajewski, P. Themann, and L. Eckstein, "Decoupled cooperative trajectory optimization for connected highly automated vehicles at urban intersections," in *Proc. IEEE Intell. Vehicles Symp. (IV)*, Jun. 2016, pp. 741–746.
- [26] P. Dai, K. Liu, Q. Zhuge, E. H.-M. Sha, V. C. S. Lee, and S. H. Son, "Quality-of-experience-oriented autonomous intersection control in vehicular networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 17, no. 7, pp. 1956–1967, Jul. 2016.
- [27] I. Lamouik, A. Yahyaoui, and M. A. Sabri, "Smart multi-agent traffic coordinator for autonomous vehicles at intersections," in *Proc. Int. Conf. Adv. Technol. Signal Image Process. (ATSIP)*, May 2017, pp. 1–6.
- [28] M. Hausknecht, T.-C. Au, and P. Stone, "Autonomous intersection management: Multi-intersection optimization," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, Sep. 2011, pp. 4581–4586.
- [29] C. Wuthishuwong and A. Traechtler, "Vehicle to infrastructure based safe trajectory planning for autonomous intersection management," in *Proc. 13th Int. Conf. ITS Telecommun. (ITST)*, Nov. 2013, pp. 175–180.
- [30] M. O. Sayin, C.-W. Lin, S. Shiraishi, J. Shen, and T. Basar, "Information-driven autonomous intersection control via incentive compatible mechanisms," *IEEE Trans. Intell. Transp. Syst.*, vol. 20, no. 3, pp. 912–924, Mar. 2019.
- [31] M. Choi, A. Rubenecia, and H. H. Choi, "Reservation-based intersection crossing scheme for autonomous vehicles traveling in a speed range," in *Proc. Int. Conf. Inf. Netw. (ICOIN)*, Jan. 2019, pp. 84–87.
- [32] M. Choi, A. Rubenecia, and H. H. Choi, "Reservation-based autonomous intersection management considering vehicle failures in the intersection," in *Proc. Int. Conf. Inf. Netw. (ICOIN)*, Jan. 2020, pp. 654–659.
- [33] A. Sahba, R. Sahba, P. Rad, and M. Jamshidi, "Optimized IoT based decision making for autonomous vehicles in intersections," in *Proc. IEEE 10th Annu. Ubiquitous Comput., Electron. Mobile Commun. Conf. (UEMCON)*, Oct. 2019, pp. 0203–0206.
- [34] D. Miculescu and S. Karaman, "Polling-systems-based autonomous vehicle coordination in traffic intersections with no traffic signals," *IEEE Trans. Autom. Control*, vol. 65, no. 2, pp. 680–694, Feb. 2020.
- [35] K. Zhang, Y. Mao, S. Leng, Y. He, and Y. Zhang, "Mobile-edge computing for vehicular networks: A promising network paradigm with predictive offloading," *IEEE Veh. Technol. Mag.*, vol. 12, no. 2, pp. 36–44, Jun. 2017.
- [36] A. Giridhar and P. R. Kumar, "Scheduling automated traffic on a network of roads," *IEEE Trans. Veh. Technol.*, vol. 55, no. 5, pp. 1467–1474, Sep. 2006.
- [37] J. Kennedy and R. Eberhart, "Particle swarm optimization," in *Proc. IEEE ICNN*, vol. 4, Nov./Dec. 1995, pp. 1942–1948.
- [38] M. A. Montes de Oca, T. Stutzle, M. Birattari, and M. Dorigo, "Frankenstein's PSO: A composite particle swarm optimization algorithm," *IEEE Trans. Evol. Comput.*, vol. 13, no. 5, pp. 1120–1132, Oct. 2009.
- [39] E. Alba, J. Garcia-Nieto, L. Jourdan, and E.-G. Talbi, "Gene selection in cancer classification using PSO/SVM and GA/SVM hybrid algorithms," in *Proc. IEEE Congr. Evol. Comput.*, Sep. 2007, pp. 284–290.
- [40] P. N. Suganthan, "Particle swarm optimiser with neighbourhood operator," in *Proc. Congr. Evol. Comput.-CEC*, Jul. 1999, pp. 1958–1962.
- [41] D. I. Robertson, "TRANSYT: Traffic network study tool," in *Proc. 4th Int. Symp. Theory Traffic Flow*, Karlsruhe, Germany, 1968, pp. 22–41.
- [42] C. Daganzo, *Fundamentals of Transportation and Traffic Operations*, vol. 30. Oxford, U.K.: Pergamon, 1997.



YUHENG ZHANG received the B.S. degree in communication engineering from the Beijing University of Post and Telecommunication (BUPT), Beijing, China, where he is currently pursuing the M.E. degree in electronic and information engineering. His research interests include edge computing, autonomous intersection control, and vehicular ad-hoc networks (VANETs).



LUNING LIU received the B.S. degree in communication engineering from Beijing Jiaotong University, Beijing, China, in 2017. She is currently pursuing the Ph.D. degree in communications engineering with the Beijing University of Posts and Telecommunications, Beijing. She is also with the Beijing Key Laboratory of Network System Architecture and Convergence, Beijing. Her current research interests include connected vehicles and mobile-edge computing.



ZHAOMING LU (Member, IEEE) received the B.S. degree in communication engineering from Zhengzhou University, in 2007, and the Ph.D. degree in information and communication engineering from the Beijing University of Posts and Telecommunications, in 2012. He is currently a Faculty Member with the Beijing University of Posts and Telecommunications. His research interests are in the areas of opensource 5G, network aided autonomous driving, WiFi sensing and imaging.

He has served as the general chair, a TPC Chair, and a TPC Member for several international workshops and conferences, and he is on the editorial board of two international journals. He has got the best paper award of IEEE ICEI 2018, and the best demo of IEEE ISWCS 2016.



LUHAN WANG (Member, IEEE) received the Ph.D. degree in communications engineering from the Beijing University of Posts and Telecommunications, China. His current research interests include network architecture, network function virtualization, software-defined networks, and study of implementing NR CN for open air interface.



XIANGMING WEN (Senior Member, IEEE) received the B.E., M.S., and Ph.D. degrees in electrical engineering from the Beijing University of Posts and Telecommunications (BUPT), Beijing, China. He is currently the Vice President of BUPT, where he is also a Professor with the Communication Network Center and the Director of the Beijing Key Laboratory of Network System Architecture and Convergence. He has authored more than 100 published articles. He is the Principle

Investigator of more than 18 projects, including the National Key Project of Hi-Tech Research and Development Program of China (863 Program) and the National Natural Science Foundation of China. His current research interests include open source 5G, broadband mobile communication theory, multimedia communications, and information processing. He is the Vice Director of the Organization Committee of the China Telecommunication Association.

...